Human Activity Recognition Using Feature Engineering + Machine Learning

Week■1 Deliverable Report

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1. Dataset Summary

This project uses the UCI Human Activity Recognition (HAR) dataset collected using smartphone sensors. Data was recorded from subjects performing six activities using tri■axial accelerometer and gyroscope sensors. Signals were sampled at 50 Hz and segmented into fixed windows of 128 samples. The task is formulated as a multiclass classification problem.

Property	Value
Total Samples	10,299
Training Samples	7,352
Testing Samples	2,947
Number of Activities	6
Sensors Used	Accelerometer, Gyroscope
Problem Type	Multiclass Classification

2. Feature Engineering

Custom features were extracted directly from raw sensor signals to make them suitable for classical machine learning algorithms.

Category	Features Used
Statistical	Mean, Std, Min, Max, Skewness, Kurtosis
Temporal	SMA, Energy, ZCR, Peak■to■Peak
Frequency	Spectral Entropy (FFT based)

3. Machine Learning Models

The following classical machine learning models were trained using standardized features and an 80/20 train■validation split.

Model	Description
Logistic Regression	Linear baseline classifier
Support Vector Machine	Non■linear classification using kernel
Random Forest	Ensemble of decision trees
Naive Bayes	Probabilistic generative model

4. Evaluation Metrics

Model performance was evaluated using standard classification metrics including Accuracy, Precision, Recall, F1■Score, and Confusion Matrix.

5. Results and Discussion

The Random Forest classifier achieved the best overall performance. Walking■related activities were classified accurately due to strong motion patterns. Confusion mainly occurred between static activities such as sitting, standing, and laying.

6. Limitations and Future Work

Classical machine learning relies on handcrafted features and may miss temporal dependencies. Future work will focus on deep learning models such as CNNs and LSTMs to learn features directly from raw sensor data.

7. Conclusion

This work demonstrates an effective feature engineering and classical machine learning pipeline for human activity recognition. The results establish a strong baseline for future deep learning approaches.